QUALIFICATION
REQUIREMENTS
REMOVED

MIL-M-38510/476A(NASA)
9 August 1983
SUPERSEDING
MIL-M-38510/476(NASA)
30 September 1980

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS 4-BIT BUS BUFFER/SEPARATOR, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by George C. Marshall Space Flight Center, National Aeronautics and Space Administration, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic silicon, silicon gate CMOS microcircuit. One product assurance class and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be as specified in MIL-M-38510, except the "JAN" and "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-bit memory data bus buffer/separator
02	4-bit Input/Output bus buffer/separator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Storage temperature range (T_{stg})	-65°C to +150°C
Case operating temperature range (T_C)	-55°C to +125°C
Supply voltage range (V_{DD} - V_{SS})	-0.5 V dc to +13 V dc
Maximum power dissipation per package (P_D):	
For T_C = -55 to +100°C	500 mW
For T_C = +100 to +125°C	Derate linearly at 12 mW/°C to 200 mW
Device dissipation per output transistor:	
For T_C = -55°C to +125°C	100 mW
Input voltage range, all inputs	-0.5 V dc to V_{DD} +0.5 V dc
DC input current (any one input)	±10 mA
Lead temperature (soldering, 10 seconds):	
At distance 1/16 ±1/32 inch (1.59 ±0.79 mm) from case	265°C
Thermal resistance, junction to case (θ_{JC}):	
Case E	0.0833°C/mW
Case F	0.15°C/mW
Junction temperature (T_J)	+150°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C. Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN: EG02, Marshall Space Flight Center, AL 35812, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{DD} - V_{SS})	+4.5 V dc minimum to +11 V dc maximum
Input low (V_{IL}) voltage range	0 to +1.5 V dc at V_{DD} = +5 V dc; 0 to +3 V dc at V_{DD} = +10 V dc
Input high (V_{IH}) voltage range	+3.5 V dc to +5 V dc at V_{DD} = +5 V dc; +7 V dc to +10 V dc at V_{DD} = +10 V dc

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagram. The logic diagram shall be as specified on figure 2.

3.2.3 Truth tables and functional description. The functional description shall be as specified in the truth table on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in MIL-M-38510 and 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. Electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS} = 0 V, -55°C ≤ T_A ≤ 125°C Unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD}	$V_{IC(POS)}$	T_C = 25°C, V_{DD} = 0 V, V_{SS} = Open, Output = Open, I_I = 1 mA		All		3.0	V
Negative clamping input to V_{SS}	$V_{IC(NEG)}$	T_C = 25°C, V_{DD} = Open, V_{SS} = 0 V, Output = Open, I_I = -1 mA		All		-3.0	V
Quiescent supply current	I_{SS}	Any combination of inputs	V_{DD} = 13.0 V	All		-100	µA
High level output voltage	V_{OH1}	V_{DD} = 10 V, I_{OH} ≤ 1 µA (see table III for V_I)		All	9.95		V
Low level output voltage	V_{OL1}	V_{DD} = 10.0 V, I_{OL} ≤ 1 µA (see table III for V_I)		All		0.05	V
Output high (source) current	I_{OH1} I_{OH2}	V_{DD} = 5.0 V, V_{OH1} = 4.6 V V_{DD} = 10.0 V, V_{OH2} = 9.5 V		All	-1.1 -2.6		mA
Output low (sink) current	I_{OL1} I_{OL2}	V_{DD} = 5.0 V, V_{OL1} = 0.4 V V_{DD} = 10.0 V, V_{OL2} = 0.5 V		All	1.5 2.6		mA
High impedance three-state output leakage	I_Z (high Z)	V_{DD} = 13.0 V (see table III for V_I)		All		±5	µA
Input current, high	I_{IH}	V_{DD} = 13.0 V, V_I = 13.0 V		All		1	µA
Input current, low	I_{IL}	V_{DD} = 13.0 V, V_I = 0		All		-1	µA
Propagation delay high to low	t_{PHL}	C_L = 100 pF, R_L = 200 kΩ	V_{DD} = 5.0 V	All	150	300	ns
			V_{DD} = 10.0 V	All	9	180	ns
Propagation delay low to high	t_{PLH}	C_L = 100 pF, R_L = 100 kΩ	V_{DD} = 5.0 V	All	15	300	ns
			V_{DD} = 10.0 V	All	9	180	ns
Propagation delay high Z to low level	t_{PZL}	C_L = 100 pF, R_L = 10 kΩ	V_{DD} = 5.0 V	All	15	300	ns
			V_{DD} = 10.0 V	All	9	180	ns
Propagation delay low Z to high level	t_{PZH}	C_L = 100 pF, R_L = 10 kΩ	V_{DD} = 5.0 V	All	15	300	ns
			V_{DD} = 10.0 V	All	9	180	ns
Transition time high to low level (DB or DO)	t_{THL}	C_L = 100 pF, R_L = 200 kΩ	V_{DD} = 5.0 V	All		156	ns
			V_{DD} = 10.0 V	All		74	ns
Transition time low to high level (DB or DO)	t_{TLH}	C_L = 100 pF, R_L = 200 kΩ	V_{DD} = 5.0 V	All		156	ns
			V_{DD} = 10.0 V	All		74	ns

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3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line, not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (table III) unless otherwise specified
	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1
Interim electrical parameters (between static and dynamic) (method 5004)	N/A
Final electrical test parameters (method 5004)	1*,2,7*,8
Group A test requirements (method 5005)	1,2,7,8,9
Group B electrical parameters (method 5005)	None
Group B V_{ZAP} (method 5005)	4.5.2
Group C end-point electrical parameters (method 5005)	1,2,7,8
Additional electrical subgroups for group C periodic inspections (method 5005)	10,11
Group D end-point electrical parameters (method 5005)	1,2,7,8

* PDA applies to subgroups 1 and 7 (see 4.2b).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following modifications and additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) The burn-in test as described in 3.1.10 of method 5004 shall be performed as indicated. (3.1.11 and 3.1.12 are not required; 3.1.9 is required for class B devices.)

(2) For class B devices, dynamic burn-in shall be performed for 160 hours minimum at 125°C ambient temperature, using the circuit shown on figure 4, or equivalent.

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b. The percent defective allowable (PDA) for class B devices shall be 10 percent (or one device, whichever is greater) based on failures from group A, subgroups 1 and 7 (after cooldown) of final electrical test, excluding delta limits, in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

c. Interim and final electrical parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. A special subgroup shall be added using a LTPD of 15 for class B. This subgroup shall consist of a high-voltage test of the input protection circuits, V_{ZAP} (see 4.5.2).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C, minimum.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

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Device types 01 and 02

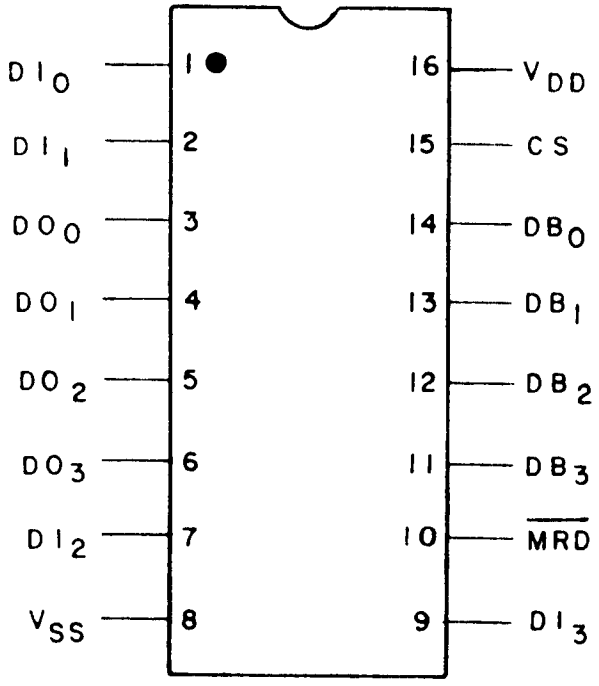


FIGURE 1. Terminal connections.

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Device type 01

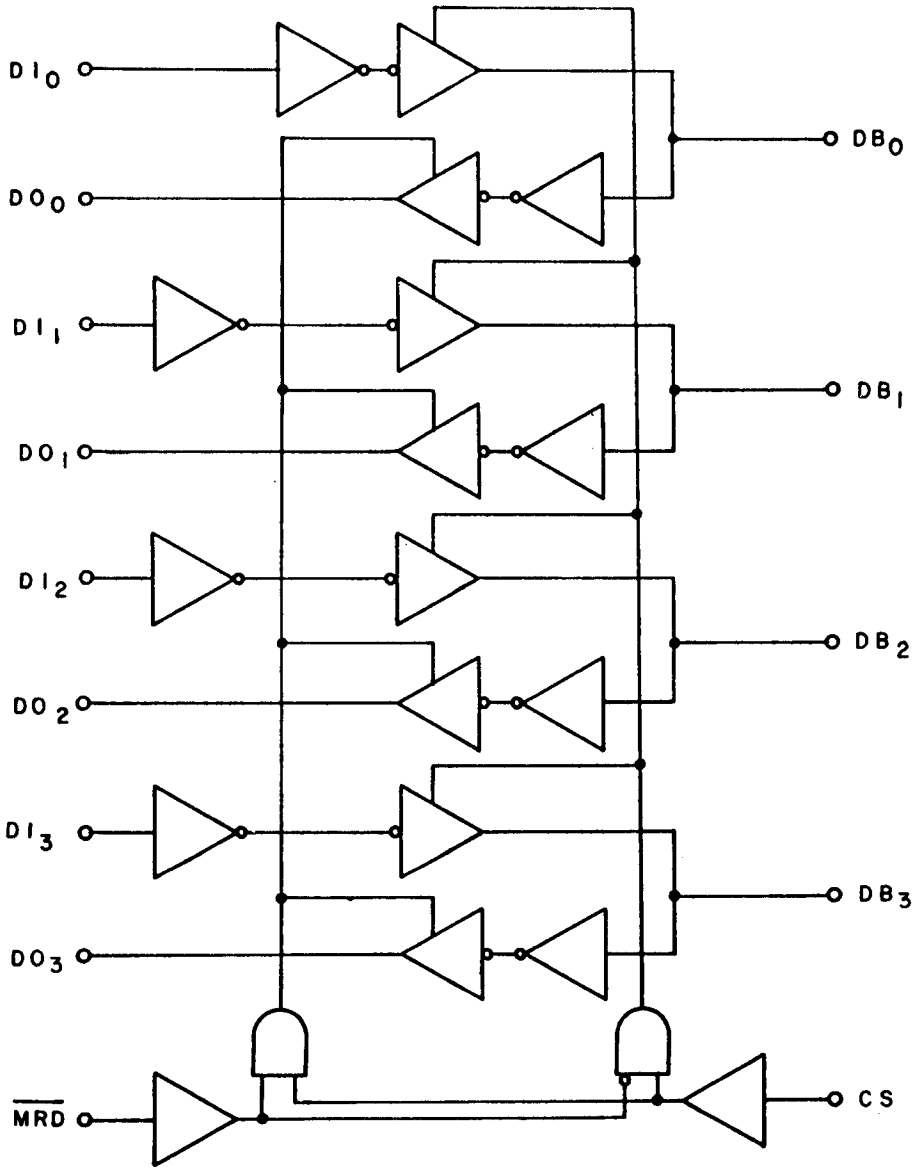


FIGURE 2. Logic diagram.

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Device type 02

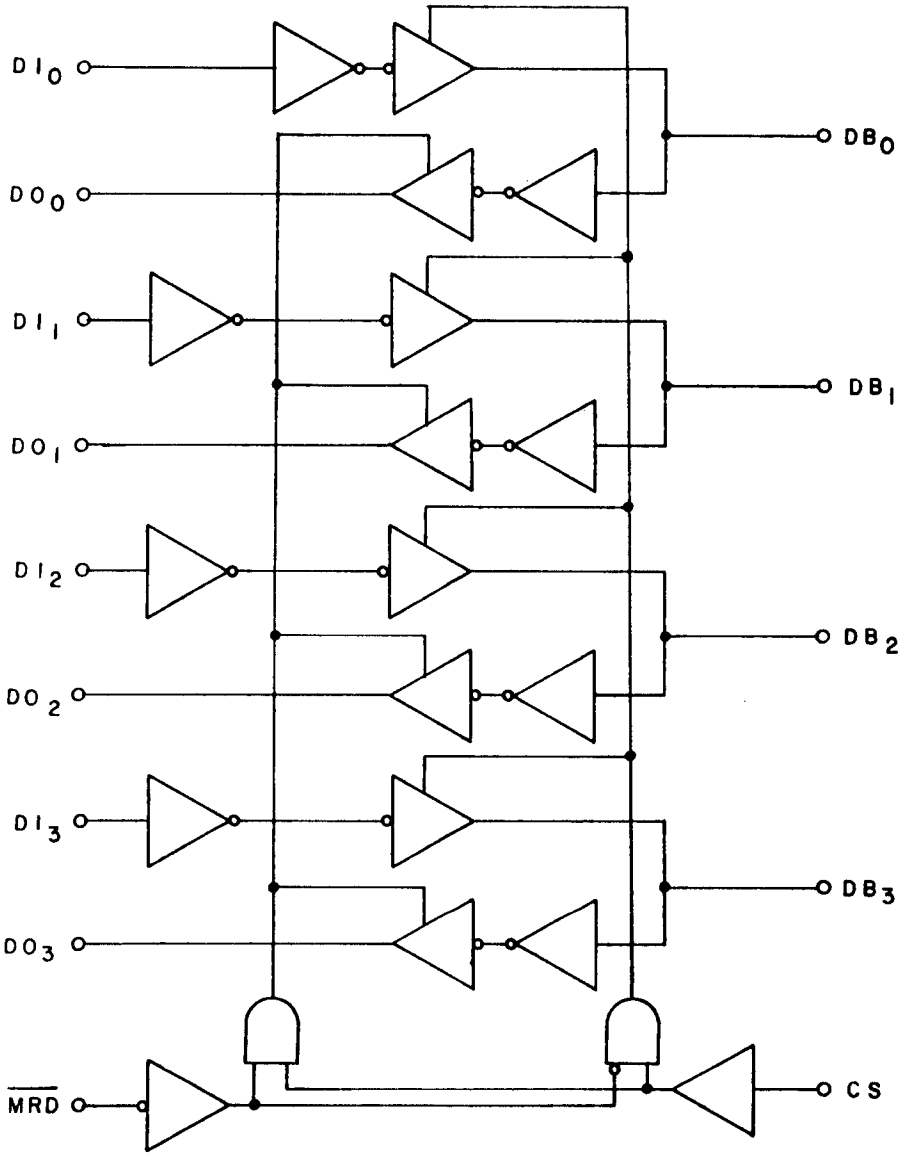


FIGURE 2. Logic diagram - Continued.

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Device type 01

CS	$\overline{MRD}$	DATA BUS OUTPUTS (DB 0 - DB 3)	DATA OUTPUTS (DO 0 - DO 3)
L	X	High impedance	High impedance
H	L	Data input	High impedance
H	H	High impedance	Data bus input

Device type 02

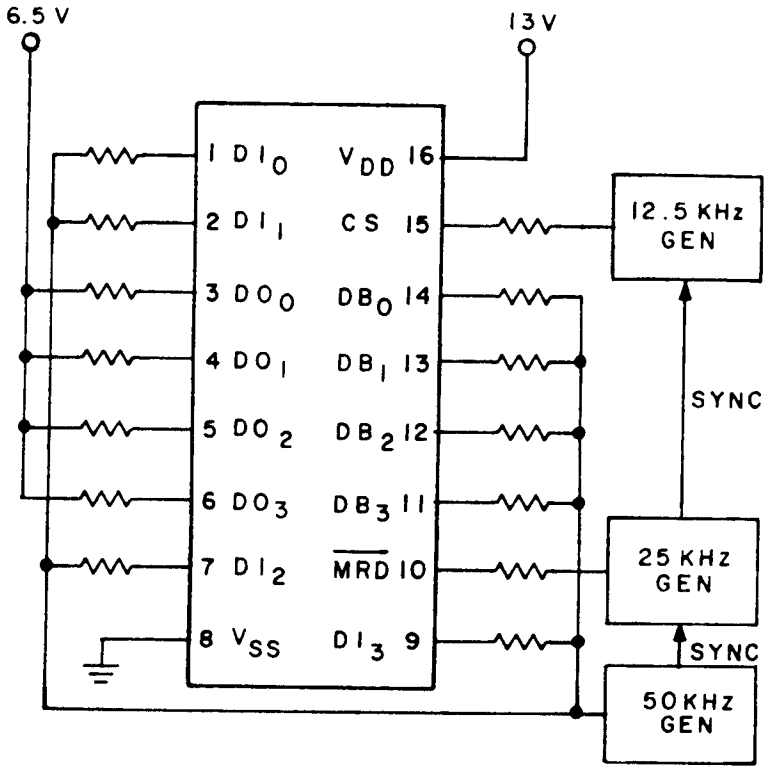
CS	$\overline{MRD}$	DATA BUS OUTPUTS (DB 0 - DB 3)	DATA OUTPUTS (DO - 0 DO 3)
L	X	High impedance	High impedance
H	L	High impedance	Data bus input
H	H	Data input	High impedance

NOTES:
1. L = Low level logic
2. H = High level logic
3. X = Don't care

FIGURE 3. Truth table.

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Device types 01 and 02



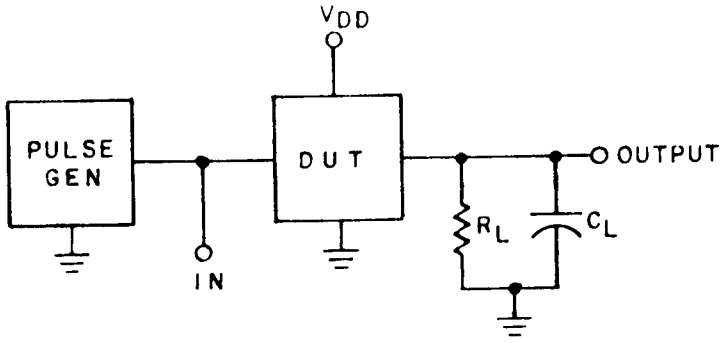
NOTES:
1. All resistor shall be 0-47 kilohms maximum.
2. Input signal requirements:

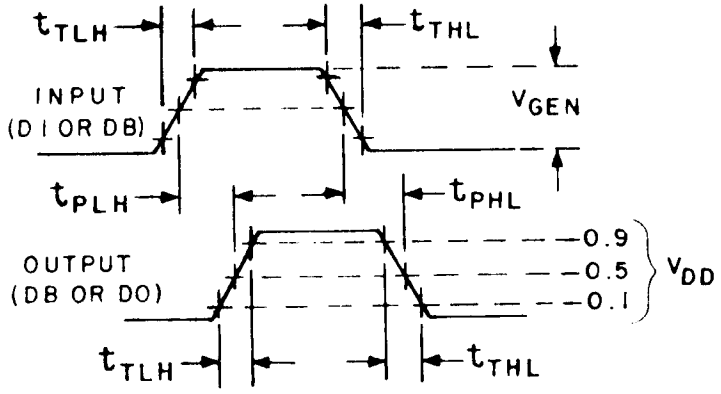
   a. Square wave, 50% duty cycle synchronized like a 3-stage divider.
   b. Voltage = 0 to 13 V ±2%.
   c. $t_{TLH} = t_{THL} \leq 1$ µs.

FIGURE 4. Dynamic burn-in and steady state life test circuit.

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Device types 01 and 02





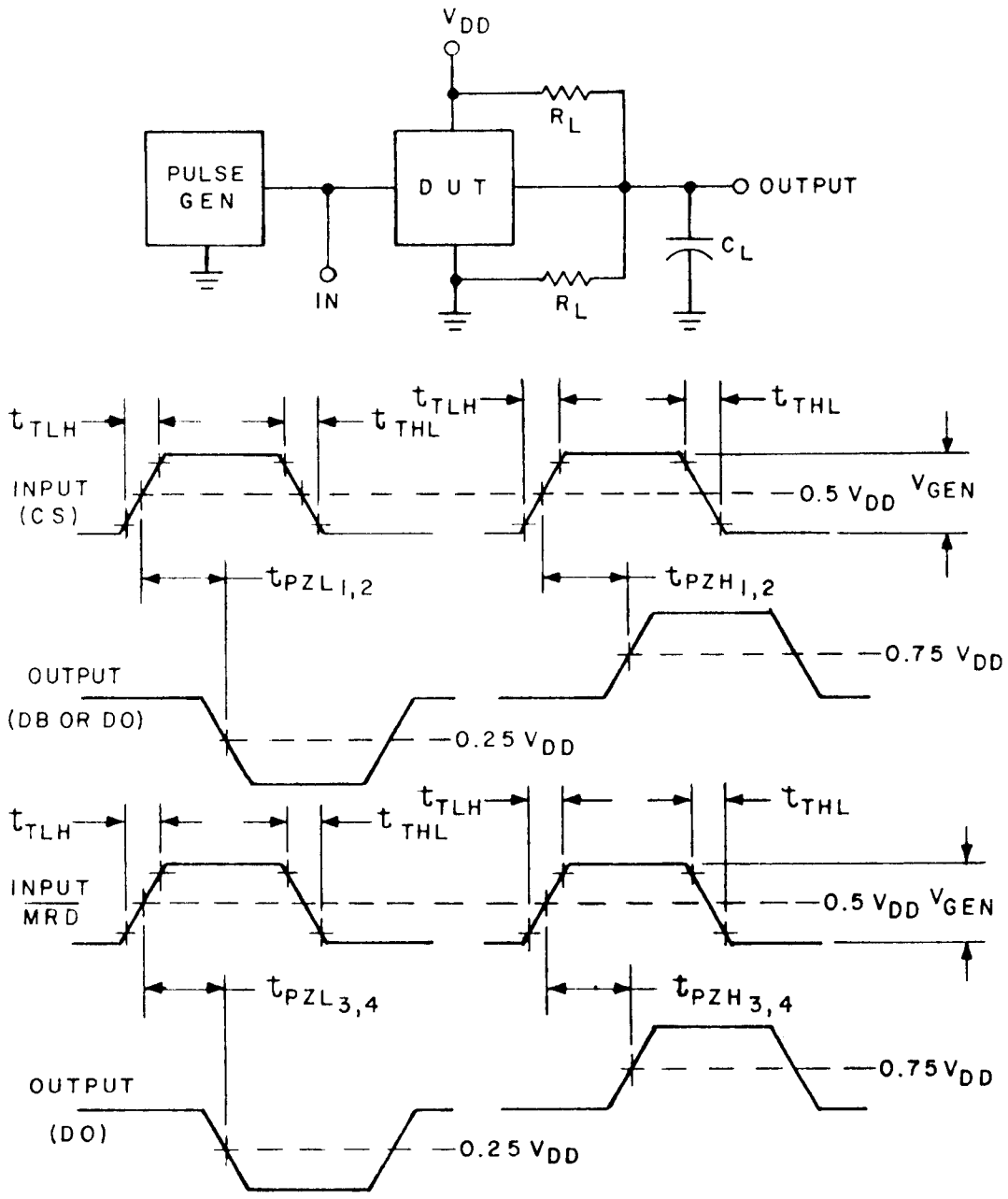
R_L = 200 kΩ ±1%.
C_L = 100 pF ±5% (includes wiring and probe capacitance).

NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD}$ ±1.0%, $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. See table III for complete terminal connections.

FIGURE 5a. Switching time test circuit and waveforms.

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Device types 01 and 02



R_L = 10 kΩ ±1%.
C_L = 100 pF ±5% (including wiring and probe capacitance).

NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD}$ ±1.0%, $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. See table III for complete terminal conditions.

FIGURE 5b. Switching time test circuit and waveforms.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	CS	V_{DD}		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1 mA															GND	DI_0		3.0					V
		2		1 mA														"	DI_1		"					"
		3							1 mA									"	DI_2		"					"
		4									1 mA							"	DI_3		"					"
		5										1 mA						"	$\overline{MRD}$		"					"
		6											1 mA					"	DB_3		"					"
		7												1 mA				"	DB_2		"					"
		8													1 mA			"	DB_1		"					"
		9														1 mA		"	DB_0		"					"
		10										1					1 mA	"	CS		"					"
V_{IC} (neg)		11	-1 mA							GND									DI_0		-3.0					V
		12		-1 mA						"									DI_1		"					"
		13							-1 mA	"									DI_2		"					"
		14								"	-1 mA								DI_3		"					"
		15								"		-1 mA							$\overline{MRD}$		"					"
		16								"			-1 mA						DB_3		"					"
		17								"				-1 mA					DB_2		"					"
		18								"					-1 mA				DB_1		"					"
		19								"						-1 mA			DB_0		"					"
		20								"							-1 mA		CS		"					"
I_{SS} 2/	3005	21	13.0 V	13.0 V					13.0 V	"	13.0 V	GND	GND	GND	GND	GND	GND	13.0 V	V_{SS}		20		100		20	uA
	"	22	"	"					"	"	"	13.0 V	"	"	"	"	GND	"	"		"		"		"	"
	"	23	"	"					"	"	"	"	13.0 V	13.0 V	13.0 V	13.0 V	13.0 V	"	"		"		"		"	"
	"	24	"	"					"	"	"	GND	"	"	"	"	"	"	"		"		"		"	"
	"	25	"	"					"	"	"	GND	"	"	"	"	"	"	"		"		"		"	"
	"	26	GND	GND					GND	"	GND	GND	"	"	"	"	"	"	"		"		"		"	"
V_{OH1}	3006	27	10.0 V	10.0 V					10.0 V	"	10.0 V	10.0 V	10.0 V	10.0 V	10.0 V	10.0 V	10.0 V	10.0 V	DO_0	9.95		9.95		9.95		V
	"	28	"	"					"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
	"	29	"	"					"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
	"	30	"	"					"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
	"	31	"	"					"	"	"	"	"	"	"	"	"	"	DB_3	"		"		"		"
	"	32	"	"					"	"	"	"	"	"	"	"	"	"	DB_2	"		"		"		"
	"	33	"	"					"	"	"	"	"	"	"	"	"	"	DB_1	"		"		"		"
	"	34	"	"					"	"	"	GND	"	"	"	"	"	"	DB_0	"		"		"		"
V_{OL1}	3007	35	GND	GND					GND	"	GND	10.0 V	GND	GND	GND	GND	"	"	DO_0		.05		.05		.05	V
	"	36	"	"					"	"	"	"	"	"	"	"	"	"	DO_1		"		"		"	"
	"	37	"	"					"	"	"	"	"	"	"	"	"	"	DO_2		"		"		"	"
	"	38	"	"					"	"	"	"	"	"	"	"	"	"	DO_3		"		"		"	"
	"	39	"	"					"	"	"	"	"	"	"	"	"	"	DB_3		"		"		"	"
	"	40	"	"					"	"	"	"	"	"	"	"	"	"	DB_2		"		"		"	"
	"	41	"	"					"	"	"	"	"	"	"	"	"	"	DB_1		"		"		"	"
	"	42	"	"					"	"	"	GND	"	"	"	"	"	"	DB_0		"		"		"	"
I_{OH1}		43	V_{DD}	V_{DD}	4.6 V	4.6 V	4.6 V	4.6 V	V_{DD}	"	V_{DD}	V_{DD}	V_{DD}	V_{DD}	V_{DD}	V_{DD}	V_{DD}	5.0 V	DO_0	-1.5		-1.1		-2.1		mA
		44	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		45	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		46	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		47	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DB_3	"		"		"		"
		48	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DB_2	"		"		"		"
		49	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DB_1	"		"		"		"
		50	"	"	"	"	"	"	"	"	"	GND	"	"	"	"	"	"	DB_0	"		"		"		"
I_{OH2}		51	"	"	9.5 V	9.5 V	9.5 V	9.5 V	"	"	"	V_{DD}	V_{DD}	V_{DD}	V_{DD}	V_{DD}	"	10.0 V	DO_0	-3.5		-2.6		-4.9		"
		52	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		53	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		54	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		55	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DB_3	"		"		"		"
		56	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DB_2	"		"		"		"
		57	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DB_1	"		"		"		"
		58	"	"	"	"	"	"	"	"	"	GND	"	"	"	"	"	"	DB_0	"		"		"		"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	CS	V_{DD}		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
I_{OL1}		59	GND	GND	0.4 V				GND	GND	GND	V_{DD}	GND	GND	GND	GND	V_{DD}	5.0 V	DO_0	2.1		1.5		2.8		mA
		60	"	"		0.4 V			"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		61	"	"			0.4 V		"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		62	"	"				0.4 V	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		63	"	"					"	"	"	GND	0.4 V				"	"	DB_3	"		"		"		"
		64	"	"					"	"	"	"		0.4 V			"	"	DB_2	"		"		"		"
		65	"	"					"	"	"	"			0.4 V		"	"	DB_1	"		"		"		"
		66	"	"					"	"	"	"				0.4 V	"	"	DB_0	"		"		"		"
I_{OL2}		67	"	"	0.5 V				"	"	"	V_{DD}	GND	GND	GND	GND	"	10.0 V	DO_0	3.5		2.6		4.7		"
		68	"	"		0.5 V			"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		69	"	"			0.5 V		"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		70	"	"				0.5 V	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		71	"	"					"	"	"	GND	0.5 V				"	"	DB_3	"		"		"		"
		72	"	"					"	"	"	"		0.5 V			"	"	DB_2	"		"		"		"
		73	"	"					"	"	"	"			0.5 V		"	"	DB_1	"		"		"		"
		74	"	"					"	"	"	"				0.5 V	"	"	DB_0	"		"		"		"
I_{ZH} 2/ (HI Z)		75	"	"					"	"	"	"	11.0 V				GND	13.0 V	DB_3		1		5		1	μA
		76	"	"					"	"	"	"		11.0 V			"	"	DB_2		"		"		"	"
		77	"	"					"	"	"	"			11.0 V		"	"	DB_1		"		"		"	"
		78	"	"					"	"	"	"				11.0 V	"	"	DB_0		"		"		"	"
I_{ZL} 2/ (HI Z)		79	11.0 V	11.0 V					11.0 V	"	11.0 V	"	GND				"	"	DB_3		-1		-5		-1	"
		80	"	"					"	"	"	"		GND			"	"	DB_2		"		"		"	"
		81	"	"					"	"	"	"			GND		"	"	DB_1		"		"		"	"
		82	"	"					"	"	"	"				GND	"	"	DB_0		"		"		"	"
I_{IH} 2/	3010	83	13.0 V	GND					GND	"	GND	"	GND	GND	GND	"	"	"	DI_0		.5		1		.5	"
	"	84	GND	13.0 V					GND	"	"	"	"	"	"	"	"	"	DI_1		"		"		"	"
	"	85	"	GND					13.0 V	"	"	"	"	"	"	"	"	"	DI_2		"		"		"	"
	"	86	"	"					GND	"	13.0 V	"	"	"	"	"	"	"	DI_3		"		"		"	"
	"	87	"	"					"	"	"	13.0 V	"	"	"	"	"	"	$\overline{MRD}$		"		"		"	"
	"	88	"	"					"	"	"	"	"	"	"	"	13.0 V	"	CS		"		"		"	"
I_{IL} 2/	3009	89	GND	13.0 V					13.0 V	"	"	"	13.0 V	13.0 V	13.0 V	13.0 V	"	"	DI_0		-.5		-1		-.5	"
	"	90	13.0 V	GND					13.0 V	"	"	"	"	"	"	"	"	"	DI_1		"		"		"	"
	"	91	"	13.0 V					GND	"	"	"	"	"	"	"	"	"	DI_2		"		"		"	"
	"	92	"	"					13.0 V	"	GND	"	"	"	"	"	"	"	DI_3		"		"		"	"
	"	93	"	"					"	"	13.0 V	GND	"	"	"	"	GND	"	$\overline{MRD}$		"		"		"	"
	"	94	"	"					"	"	13.0 V	13.0 V	"	"	"	"	GND	"	CS		"		"		"	"

				Subgroup 7 T_C = 25°C		Subgroup 8				
						T_C = 125°C		T_C = -55°C		
				Min	Max	Min	Max	Min	Max	
Functional tests	3014	95	3/	3/	3/	3/	3/	3/	3/	
	3014	96	3/	3/	3/	3/	3/	3/	3/	

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	CS	V_{DD}		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
Functional tests	3014	95	3/	3/	3/	3/	3/	3/	3/	GND	3/	3/	3/	3/	3/	3/	3/	4.5 V	All outputs	3/	3/	3/	3/	3/	3/	
	3014	96	3/	3/	3/	3/	3/	3/	3/	GND	3/	3/	3/	3/	3/	3/	3/	13.0 V	All outputs	3/	3/	3/	3/	3/	3/	

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/																Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	$\overline{CS}$	V_{DD}		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig. 5a	97	IN	GND					GND	GND	GND	GND				OUT	V_{DD}	5.0 V	DI_0 to DB_0	9	190	15	300	9	190	ns
	"	98	GND	IN					GND	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	99	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	100	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	101	"	"	OUT				"	"	GND	V_{DD}				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	102	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	103	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
	"	104	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"
t_{PHL2}	"	105	IN	GND					"	"	"	GND				OUT	"	10.0 V	DI_0 to DB_0	5	100	9	180	5	100	"
	"	106	GND	IN					"	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	107	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	108	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	109	"	"	OUT				"	"	GND	V_{DD}				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	110	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	111	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
	"	112	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"
t_{PLH1}	"	113	IN						"	"	"	GND				OUT	"	5.0 V	DI_0 to DB_0	9	190	15	300	9	190	"
	"	114	GND	IN					"	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	115	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	116	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	117	"	"	OUT				"	"	GND	V_{DD}				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	118	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	119	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
	"	120	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"
t_{PLH2}	"	121	IN	"					"	"	"	GND				OUT	"	10.0 V	DI_0 to DB_0	5	100	9	180	5	100	"
	"	122	GND	IN					"	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	123	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	124	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	125	"	"	OUT				"	"	GND	V_{DD}				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	126	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	127	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
		128	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/																Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	CS	V_{DD}		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PZL1}	Fig. 5b	129	GND	GND					GND	GND	GND	GND				OUT	IN	5.0 V	CS to DB_0	10	200	15	300	10	200	ns
	"	130	"	"					"	"	"	"			OUT		"	"	CS to DB_1	"	"	"	"	"	"	"
	"	131	"	"					"	"	"	"		OUT			"	"	CS to DB_2	"	"	"	"	"	"	"
	"	132	"	"					"	"	"	"	OUT				"	"	CS to DB_3	"	"	"	"	"	"	"
	"	133	"	"	OUT				"	"	"	V_{DD}	GND	GND	GND	GND	"	"	CS to DO_0	"	"	"	"	"	"	"
	"	134	"	"		OUT			"	"	"	"				"	"	"	CS to DO_1	"	"	"	"	"	"	"
	"	135	"	"			OUT		"	"	"	"				"	"	"	CS to DO_2	"	"	"	"	"	"	"
	"	136	"	"				OUT	"	"	"	"				"	"	"	CS to DO_3	"	"	"	"	"	"	"
t_{PZL2}	Fig. 8b	137	"	"					"	"	"	GND				OUT	"	10.0 V	CS to DB_0	5	100	9	180	5	100	"
	"	138	"	"					"	"	"	"			OUT		"	"	CS to DB_1	"	"	"	"	"	"	"
	"	139	"	"					"	"	"	"		OUT			"	"	CS to DB_2	"	"	"	"	"	"	"
	"	140	"	"					"	"	"	"	OUT				"	"	CS to DB_3	"	"	"	"	"	"	"
	"	141	"	"	OUT				"	"	"	V_{DD}	GND	GND	GND	GND	"	"	CS to DO_0	"	"	"	"	"	"	"
	"	142	"	"		OUT			"	"	"	"	"	"	"	"	"	"	CS to DO_1	"	"	"	"	"	"	"
		143	"	"			OUT		"	"	"	"	"	"	"	"	"	"	CS to DO_2	"	"	"	"	"	"	"
		144	"	"				OUT	"	"	"	"	"	"	"	"	"	"	CS to DO_3	"	"	"	"	"	"	"
t_{PZL3}		145	"	"	OUT				"	"	"	IN	"	"	"	"	V_{DD}	5.0 V	$\overline{MRD}$ to DO_0	10	200	15	300	10	200	"
		146	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
		147	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
		148	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{PZL4}		149	"	"	OUT				"	"	"	"	"	"	"	"	"	10.0 V	$\overline{MRD}$ to DO_0	5	100	9	180	5	100	"
		150	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
		151	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
		152	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{PZH1}		153	V_{DD}	V_{DD}					V_{DD}	"	V_{DD}	GND	GND	GND	GND	OUT	IN	5.0 V	CS to DB_0	10	200	15	300	10	200	"
		154	"	"					"	"	"	"	"	GND	OUT	GND	"	"	CS to DB_1	"	"	"	"	"	"	"
		155	"	"					"	"	"	"	"	OUT	GND	"	"	"	CS to DB_2	"	"	"	"	"	"	"
		156	"	"					"	"	"	"	OUT	GND	GND	"	"	"	CS to DB_3	"	"	"	"	"	"	"
		157	"	"	OUT				"	"	"	V_{DD}	V_{DD}	V_{DD}	V_{DD}	V_{DD}	"	"	CS to DO_0	"	"	"	"	"	"	"
		158	"	"		OUT			"	"	"	"	"	"	"	"	"	"	CS to DO_1	"	"	"	"	"	"	"
		159	"	"			OUT		"	"	"	"	"	"	"	"	"	"	CS to DO_2	"	"	"	"	"	"	"
		160	"	"				OUT	"	"	"	"	"	"	"	"	"	"	CS to DO_3	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F Test no.	Terminal conditions and limits 1/ DI_0 1	DI_1 2	DO_0 3	DO_1 4	DO_2 5	DO_3 6	DI_2 7	V_{SS} 8	DI_3 9	$\overline{MRD}$ 10	DB_3 11	DB_2 12	DB_1 13	DB_0 14	CS 15	V_{DD} 16	Measured terminal	Test limits Subgroup 9 T_C = 25°C Min	Max	Subgroup 10 T_C = 125°C Min	Max	Subgroup 11 T_C = -55°C Min	Max	Unit
t_{PZH2}	Fig. 5b	161	V_{DD}	V_{DD}					V_{DD}	GND	V_{DD}	GND				OUT	IN	10.0 V	CS to DB_0	5	100	9	180	5	100	ns
	"	162	"	"					"	"	"	"			OUT		"	"	CS to DB_1	"	"	"	"	"	"	"
	"	163	"	"					"	"	"	"		OUT			"	"	CS to DB_2	"	"	"	"	"	"	"
	"	164	"	"					"	"	"	"	OUT				"	"	CS to DB_3	"	"	"	"	"	"	"
	"	165	"	"	OUT				"	"	"	V_{DD}	V_{DD}	V_{DD}	V_{DD}	V_{DD}	"	"	CS to DO_0	"	"	"	"	"	"	"
	"	166	"	"		OUT			"	"	"	"	"	"	"	"	"	"	CS to DO_1	"	"	"	"	"	"	"
	"	167	"	"			OUT		"	"	"	"	"	"	"	"	"	"	CS to DO_2	"	"	"	"	"	"	"
	"	168	"	"				OUT	"	"	"	"	"	"	"	"	"	"	CS to DO_3	"	"	"	"	"	"	"
t_{PZH3}	"	169	"	"	OUT				"	"	"	IN	"	"	"	"	V_{DD}	5.0 V	$\overline{MRD}$ to DO_0	10	200	15	300	10	200	"
	"	170	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
	"	171	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
	"	172	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{PZH4}	"	173	"	"	OUT				"	"	"	"	"	"	"	"	"	10.0 V	$\overline{MRD}$ to DO_0	5	100	9	180	5	100	"
	"	174	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
	"	175	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
	"	176	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{THL1}	3004 Fig.5a	177	IN	GND	OUT				GND	"	GND	GND			OUT	OUT	"	5.0 V	DB_0		112		156		80	"
	"	178	GND	IN		OUT			GND	"	"	"		OUT			"	"	DB_1		"		"		"	"
	"	179	"	GND			OUT		IN	"	IN	"	OUT		OUT	IN	"	"	DB_2		"		"		"	"
	"	180	"	"				OUT	GND	"	GND	V_{DD}		IN			"	"	DB_3		"		"		"	"
	"	181	"	"					"	"	"	"	IN		IN		"	"	DO_0		"		"		"	"
	"	182	"	"					"	"	"	"					"	"	DO_1		"		"		"	"
	"	183	"	"					"	"	"	"					"	"	DO_2		"		"		"	"
	"	184	"	"					"	"	"	"					"	"	DO_3		"		"		"	"
t_{THL2}	"	185	IN	"	OUT				"	"	"	GND			OUT	OUT	"	10.0 V	DB_0		56		74		44	"
	"	186	GND	IN		OUT			"	"	"	"		OUT			"	"	DB_1		"		"		"	"
	"	187	"	GND			OUT		IN	"	IN	"	OUT		OUT	IN	"	"	DB_2		"		"		"	"
	"	188	"	"				OUT	GND	"	GND	V_{DD}		IN			"	"	DB_3		"		"		"	"
	"	189	"	"					"	"	"	"	IN		IN		"	"	DO_0		"		"		"	"
	"	190	"	"					"	"	"	"					"	"	DO_1		"		"		"	"
	"	191	"	"					"	"	"	"					"	"	DO_2		"		"		"	"
	"	192	"	"					"	"	"	"					"	"	DO_3		"		"		"	"
t_{TLH1}	"	193	IN	"	OUT				"	"	"	GND			OUT	OUT	"	5.0 V	DB_0		112		156		80	"
	"	194	GND	IN		OUT			"	"	"	"		OUT			"	"	DB_1		"		"		"	"
	"	195	"	GND			OUT		IN	"	IN	"	OUT		OUT	IN	"	"	DB_2		"		"		"	"
	"	196	"	"				OUT	GND	"	GND	V_{DD}		IN			"	"	DB_3		"		"		"	"
	"	197	"	"					"	"	"	"	IN		IN		"	"	DO_0		"		"		"	"
	"	198	"	"					"	"	"	"					"	"	DO_1		"		"		"	"
	"	199	"	"					"	"	"	"					"	"	DO_2		"		"		"	"
	"	200	"	"					"	"	"	"					"	"	DO_3		"		"		"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	$\overline{CS}$	V_{DD}		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{TLH2}	3004	201	IN	GND					GND	GND	GND	GND				OUT	V_{DD}	10.0 V	DB_0		56		74		44	ns
	Fig. 5a	202	GND	IN					GND	"	"	"			OUT		"	"	DB_1		"		"		"	"
	"	203	"	GND					IN	"	"	"		OUT			"	"	DB_2		"		"		"	"
	"	204	"	"					GND	"	IN	"	OUT				"	"	DB_3		"		"		"	"
	"	205	"	"	OUT				"	"	GND	V_{DD}				IN	"	"	DO_0		"		"		"	"
	"	206	"	"		OUT			"	"	"	"			IN		"	"	DO_1		"		"		"	"
	"	207	"	"			OUT		"	"	"	"		IN			"	"	DO_2		"		"		"	"
	"	208	"	"				OUT	"	"	"	"	IN				"	"	DO_3		"		"		"	"

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	CS	V_{DD}		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1 mA															GND	DI_0		3.0					V
		2		1 mA														"	DI_1		"					"
		3							1 mA									"	DI_2		"					"
		4									1 mA							"	DI_3		"					"
		5										1 mA						"	$\overline{MRD}$		"					"
		6											1 mA					"	DB_3		"					"
		7												1 mA				"	DB_2		"					"
		8													1 mA			"	DB_1		"					"
		9														1 mA		"	DB_0		"					"
		10															1 mA	"	CS		"					"
V_{IC} (neg)		11	-1 mA							GND									DI_0		-3.0					"
		12		-1 mA						"									DI_1		"					"
		13							-1 mA	"									DI_2		"					"
		14								"	-1 mA								DI_3		"					"
		15								"		-1 mA							$\overline{MRD}$		"					"
		16								"			-1 mA						DB_3		"					"
		17								"				-1 mA					DB_2		"					"
		18								"					-1 mA				DB_1		"					"
		19								"						-1 mA			DB_0		"					"
		20								"							-1 mA		CS		"					"
I_{SS} 2/	3005	21	13.0 V	13.0 V					13.0 V	"	13.0 V	GND	GND	GND	GND	GND	GND	13.0 V	V_{SS}		-20		-100		-20	μA
	"	22	"	"					"	"	"	13.0 V	"	"	"	"	GND	"	"		"		"		"	"
	"	23	"	"					"	"	"	"	13.0 V	13.0 V	13.0 V	13.0 V	13.0 V	"	"		"		"		"	"
	"	24	"	"					"	"	"	GND						"	"		"		"		"	"
	"	25	"	"					"	"	"	"						"	"		"		"		"	"
	"	26	GND	GND					GND	"	GND	"						"	"		"		"		"	"
V_{OH1}	3006	27	10.0 V	10.0 V					10.0 V	"	10.0 V	"	10.0 V	10.0 V	10.0 V	10.0 V	10.0 V	10.0 V	DO_0	9.95		9.95		9.95		V
	"	28	"	"					"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
	"	29	"	"					"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
	"	30	"	"					"	"	"	10.0 V	"	"	"	"	"	"	DO_3	"		"		"		"
	"	31	"	"					"	"	"	"	"	"	"	"	"	"	DB_3	"		"		"		"
	"	32	"	"					"	"	"	"	"	"	"	"	"	"	DB_2	"		"		"		"
	"	33	"	"					"	"	"	"	"	"	"	"	"	"	DB_1	"		"		"		"
	"	34	"	"					"	"	"	"	"	"	"	"	"	"	DB_0	"		"		"		"
V_{OL1}	3007	35	GND	GND					GND	"	GND	GND	GND	GND	GND	GND	"	"	DO_0		.05		.05		.05	"
	"	36	"	"					"	"	"	"	"	"	"	"	"	"	DO_1		"		"		"	"
	"	37	"	"					"	"	"	"	"	"	"	"	"	"	DO_2		"		"		"	"
	"	38	"	"					"	"	"	10.0 V	"	"	"	"	"	"	DO_3		"		"		"	"
	"	39	"	"					"	"	"	"	"	"	"	"	"	"	DB_3		"		"		"	"
	"	40	"	"					"	"	"	"	"	"	"	"	"	"	DB_2		"		"		"	"
	"	41	"	"					"	"	"	"	"	"	"	"	"	"	DB_1		"		"		"	"
	"	42	"	"					"	"	"	"	"	"	"	"	"	"	DB_0		"		"		"	"
I_{OH1}		43	V_{DD_n}	V_{DD_n}	4.6 V				V_{DD_n}	"	V_{DD_n}	GND	V_{DD}	V_{DD}	V_{DD}	V_{DD}	V_{DD}	5.0 V	DO_0	-1.5		-1.1		-2.1		mA
		44	"	"		4.6 V			"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		45	"	"			4.6 V		"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		46	"	"				4.6 V	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		47	"	"					"	"	"	V_{DD_n}	4.6 V	"	"	"	"	"	DB_3	"		"		"		"
		48	"	"					"	"	"	"	"	4.6 V	"	"	"	"	DB_2	"		"		"		"
		49	"	"					"	"	"	"	"	"	4.6 V	"	"	"	DB_1	"		"		"		"
		50	"	"					"	"	"	"	"	"	"	4.6 V	"	"	DB_0	"		"		"		"
I_{OH2}		51	"	"	9.5 V				"	"	"	GND	V_{DD}	V_{DD}	V_{DD}	V_{DD}	"	10.0 V	DO_0	-3.5		-2.6		-4.9		"
		52	"	"		9.5 V			"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		53	"	"			9.5 V		"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		54	"	"				9.5 V	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		55	"	"					"	"	"	V_{DD_n}	9.5 V	"	"	"	"	"	DB_3	"		"		"		"
		56	"	"					"	"	"	"	"	9.5 V	"	"	"	"	DB_2	"		"		"		"
		57	"	"					"	"	"	"	"	"	9.5 V	"	"	"	DB_1	"		"		"		"
		58	"	"					"	"	"	"	"	"	"	9.5 V	"	"	DB_0	"		"		"		"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits: Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		Unit
		Test no.	DI_0 1	DI_1 2	DO_0 3	DO_1 4	DO_2 5	DO_3 6	DI_2 7	V_{SS} 8	DI_3 9	$\overline{MRD}$ 10	DB_3 11	DB_2 12	DB_1 13	DB_0 14	CS 15	V_{DD} 16		Min	Max	Min	Max	Min	Max	
I_{OL1}		59	GND	GND	0.4 V				GND	GND	GND	GND	GND	GND	GND	GND	V_{DD}	5.0 V	DO_0	2.1		1.5		2.8		mA
		60	"	"		0.4 V			"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		61	"	"			0.4 V		"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		62	"	"				0.4 V	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		63	"	"					"	"	"	V_{DD}	0.4 V				"	"	DB_3	"		"		"		"
		64	"	"					"	"	"	"		0.4 V			"	"	DB_2	"		"		"		"
		65	"	"					"	"	"	"			0.4 V		"	"	DB_1	"		"		"		"
		66	"	"					"	"	"	"				0.4 V	"	"	DB_0	"		"		"		"
I_{OL2}		67	"	"	0.5 V				"	"	"	GND	GND	GND	GND	GND	"	10.0 V	DO_0	3.5		2.6		4.7		"
		68	"	"		0.5 V			"	"	"	"	"	"	"	"	"	"	DO_1	"		"		"		"
		69	"	"			0.5 V		"	"	"	"	"	"	"	"	"	"	DO_2	"		"		"		"
		70	"	"				0.5 V	"	"	"	"	"	"	"	"	"	"	DO_3	"		"		"		"
		71	"	"					"	"	"	V_{DD}	0.5 V				"	"	DB_3	"		"		"		"
		72	"	"					"	"	"	"		0.5 V			"	"	DB_2	"		"		"		"
		73	"	"					"	"	"	"			0.5 V		"	"	DB_1	"		"		"		"
		74	"	"					"	"	"	"				0.5 V	"	"	DB_0	"		"		"		"
I_{ZH} 2/ (HI Z)		75	"	"					"	"	"	GND	11.0 V				GND	13.0 V	DB_3		1		5		1	µA
		76	"	"					"	"	"	"		11.0 V			"	"	DB_2		"		"		"	"
		77	"	"					"	"	"	"			11.0 V		"	"	DB_1		"		"		"	"
		78	"	"					"	"	"	"				11.0 V	"	"	DB_0		"		"		"	"
I_{ZL} 2/ (HI Z)		79	11.0 V	11.0 V					11.0 V	"	11.0 V	"	GND				"	"	DB_3		-1		-5		-1	"
		80	"	"					"	"	"	"		GND			"	"	DB_2		"		"		"	"
		81	"	"					"	"	"	"			GND		"	"	DB_1		"		"		"	"
		82	"	"					"	"	"	"				GND	"	"	DB_0		"		"		"	"
I_{IH} 2/	3010	83	13.0 V	GND					GND	"	GND	GND	GND	GND	GND	GND	GND	"	DI_0		.5		1		.5	"
	"	84	GND	13.0 V					GND	"	"	"	"	"	"	"	"	"	DI_1		"		"		"	"
	"	85	"	GND					13.0 V	"	"	"	"	"	"	"	"	"	DI_2		"		"		"	"
	"	86	"	"					GND	"	13.0 V	"	"	"	"	"	"	"	DI_3		"		"		"	"
	"	87	"	"					"	"	GND	13.0 V	"	"	"	"	"	"	$\overline{MRD}$		"		"		"	"
	"	88	"	"					"	"	GND	"	"	"	"	"	13.0 V	"	CS		"		"		"	"
I_{IL} 2/	3009	89	"	13.0 V					13.0 V	"	13.0 V	"	13.0 V	13.0 V	13.0 V	13.0 V	"	"	DI_0		-.5		-1		-.5	"
	"	90	13.0 V	GND					13.0 V	"	"	"	"	"	"	"	"	"	DI_1		"		"		"	"
	"	91	"	13.0 V					GND	"	"	"	"	"	"	"	"	"	DI_2		"		"		"	"
	"	92	"	"					13.0 V	"	GND	"	"	"	"	"	"	"	DI_3		"		"		"	"
	"	93	"	"					"	"	13.0 V	GND	"	"	"	"	GND	"	$\overline{MRD}$		"		"		"	"
	"	94	"	"					"	"	13.0 V	13.0 V	"	"	"	"	GND	"	CS		"		"		"	"
																				Subgroup 7 T_C = 25°C		Subgroup 8 T_C = 125°C		T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Functional tests	3014	95	3/	3/	3/	3/	3/	3/	3/	GND	3/	3/	3/	3/	3/	3/	3/	4.5 V	All outputs	3/	3/	3/	3/	3/	3/	
	3014	96	3/	3/	3/	3/	3/	3/	3/	GND	3/	3/	3/	3/	3/	3/	3/	13.0 V	All outputs	3/	3/	3/	3/	3/	3/	

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test Limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	$\overline{CS}$	V_{DD}		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig. 5a	97	IN	GND					GND	GND	GND	V_{DD}				OUT	V_{DD}	5.0 V	DI_0 to DB_0	9	190	15	300	9	190	ns
	"	98	GND	IN					GND	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	99	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	100	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	101	"	"	OUT				"	"	GND	GND				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	102	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	103	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
	"	104	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"
t_{PHL2}	"	105	IN	GND					"	"	"	V_{DD}				OUT	"	10.0 V	DI_0 to DB_0	5	100	9	180	5	100	"
	"	106	GND	IN					"	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	107	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	108	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	109	"	"	OUT				"	"	GND	GND				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	110	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	111	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
	"	112	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"
t_{PLH1}	"	113	IN	"					"	"	"	V_{DD}				OUT	"	5.0 V	DI_0 to DB_0	9	190	15	300	9	190	"
	"	114	GND	IN					"	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	115	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	116	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	117	"	"	OUT				"	"	GND	GND				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	118	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	119	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
	"	120	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"
t_{PLH2}	"	121	IN	"					"	"	"	V_{DD}				OUT	"	10.0 V	DI_0 to DB_0	5	100	9	180	5	100	"
	"	122	GND	IN					"	"	"	"			OUT		"	"	DI_1 to DB_1	"	"	"	"	"	"	"
	"	123	"	GND					IN	"	"	"		OUT			"	"	DI_2 to DB_2	"	"	"	"	"	"	"
	"	124	"	"					GND	"	IN	"	OUT				"	"	DI_3 to DB_3	"	"	"	"	"	"	"
	"	125	"	"	OUT				"	"	GND	GND				IN	"	"	DB_0 to DO_0	"	"	"	"	"	"	"
	"	126	"	"		OUT			"	"	"	"			IN		"	"	DB_1 to DO_1	"	"	"	"	"	"	"
	"	127	"	"			OUT		"	"	"	"		IN			"	"	DB_2 to DO_2	"	"	"	"	"	"	"
	"	128	"	"				OUT	"	"	"	"	IN				"	"	DB_3 to DO_3	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	$\overline{CS}$	V_{DD}		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PZL1}	Fig. 5b	129	GND	GND					GND	GND	GND	V_{DD}				OUT	IN	5.0 V	CS to DB_0	10	200	15	300	10	200	ns
	"	130	"	"					"	"	"	"			OUT		"	"	$\overline{CS}$ to DB_1	"	"	"	"	"	"	"
	"	131	"	"					"	"	"	"		OUT			"	"	$\overline{CS}$ to DB_2	"	"	"	"	"	"	"
	"	132	"	"					"	"	"	"	OUT				"	"	$\overline{CS}$ to DB_3	"	"	"	"	"	"	"
	"	133	"	"	OUT				"	"	"	GND	GND	GND	GND	GND	"	"	$\overline{CS}$ to DO_0	"	"	"	"	"	"	"
		134	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_1	"	"	"	"	"	"	"
		135	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_2	"	"	"	"	"	"	"
		136	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_3	"	"	"	"	"	"	"
t_{PZL2}	"	137	"	"					"	"	"	V_{DD}	"			OUT	"	10.0 V	$\overline{CS}$ to DB_0	5	100	9	180	5	100	"
	"	138	"	"					"	"	"	"	"		OUT		"	"	$\overline{CS}$ to DB_1	"	"	"	"	"	"	"
	"	139	"	"					"	"	"	"	"	OUT			"	"	$\overline{CS}$ to DB_2	"	"	"	"	"	"	"
	"	140	"	"					"	"	"	"	OUT				"	"	$\overline{CS}$ to DB_3	"	"	"	"	"	"	"
	"	141	"	"	OUT				"	"	"	GND	GND	GND	GND	GND	"	"	$\overline{CS}$ to DO_0	"	"	"	"	"	"	"
	"	142	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_1	"	"	"	"	"	"	"
	"	143	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_2	"	"	"	"	"	"	"
	"	144	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_3	"	"	"	"	"	"	"
t_{PZL3}	"	145	"	"	OUT				"	"	"	IN	"	"	"	"	V_{DD}	5.0 V	$\overline{MRD}$ to DO_0	10	200	15	300	10	200	"
	"	146	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
	"	147	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
	"	148	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{PZL4}	"	149	"	"	OUT				"	"	"	"	"	"	"	"	"	10.0 V	$\overline{MRD}$ to DO_0	5	100	9	180	5	100	"
	"	150	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
	"	151	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
	"	152	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{PZH1}	"	153	V_{DD}	V_{DD}					V_{DD}	"	V_{DD}	V_{DD}				OUT	IN	5.0 V	CS to DB_0	10	200	15	300	10	200	"
	"	154	"	"					"	"	"	"			OUT	GND	"	"	$\overline{CS}$ to DB_1	"	"	"	"	"	"	"
	"	155	"	"					"	"	"	"		OUT	GND		"	"	$\overline{CS}$ to DB_2	"	"	"	"	"	"	"
	"	156	"	"					"	"	"	"	OUT	GND	GND		"	"	$\overline{CS}$ to DB_3	"	"	"	"	"	"	"
	"	157	"	"	OUT				"	"	"	GND	V_{DD}	V_{DD}	V_{DD}	V_{DD}	"	"	$\overline{CS}$ to DO_0	"	"	"	"	"	"	"
	"	158	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_1	"	"	"	"	"	"	"
		159	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_2	"	"	"	"	"	"	"
		160	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{CS}$ to DO_3	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases E,F	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0	DI_1	DO_0	DO_1	DO_2	DO_3	DI_2	V_{SS}	DI_3	$\overline{MRD}$	DB_3	DB_2	DB_1	DB_0	CS	V_{DD}		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PZH2}	Fig. 5b	161	V_{DD}	V_{DD}					V_{DD}	GND	V_{DD}	V_{DD}				OUT	IN	10.0 V	CS to DB_0	5	100	9	180	5	100	ns
	"	162	"	"					"	"	"	"			OUT		"	"	CS to DB_1	"	"	"	"	"	"	"
	"	163	"	"					"	"	"	"		OUT			"	"	CS to DB_2	"	"	"	"	"	"	"
	"	164	"	"					"	"	"	"	OUT				"	"	CS to DB_3	"	"	"	"	"	"	"
	"	165	"	"	OUT				"	"	"	GND	V_{DD}	V_{DD}	V_{DD}	V_{DD}	"	"	CS to DO_0	"	"	"	"	"	"	"
	"	166	"	"		OUT			"	"	"	"	"	"	"	"	"	"	CS to DO_1	"	"	"	"	"	"	"
	"	167	"	"			OUT		"	"	"	"	"	"	"	"	"	"	CS to DO_2	"	"	"	"	"	"	"
	"	168	"	"				OUT	"	"	"	"	"	"	"	"	"	"	CS to DO_3	"	"	"	"	"	"	"
t_{PZH3}	"	169	"	"	OUT				"	"	"	IN	"	"	"	"	V_{DD}	5.0 V	$\overline{MRD}$ to DO_0	10	200	15	300	10	200	"
	"	170	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
	"	171	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
	"	172	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{PZH4}	"	173	"	"	OUT				"	"	"	"	"	"	"	"	"	10.0 V	$\overline{MRD}$ to DO_0	5	100	9	180	5	100	"
	"	174	"	"		OUT			"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_1	"	"	"	"	"	"	"
	"	175	"	"			OUT		"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_2	"	"	"	"	"	"	"
	"	176	"	"				OUT	"	"	"	"	"	"	"	"	"	"	$\overline{MRD}$ to DO_3	"	"	"	"	"	"	"
t_{THL1}	3004 Fig. 5a	177	IN	GND	OUT				GND	"	GND	V_{DD}			OUT	OUT	"	5.0 V	DB_0		112		156		80	"
	"	178	GND	IN		OUT			GND	"	"	"		OUT			"	"	DB_1		"		"		"	"
	"	179	"	GND			OUT		IN	"	"	"	OUT				"	"	DB_2		"		"		"	"
	"	180	"	"				OUT	GND	"	IN	GND				IN	"	"	DB_3		"		"		"	"
	"	181	"	"					"	"	GND	"					"	"	DO_0		"		"		"	"
	"	182	"	"					"	"	"	"			IN		"	"	DO_1		"		"		"	"
	"	183	"	"					"	"	"	"		IN			"	"	DO_2		"		"		"	"
	"	184	"	"					"	"	"	"	IN				"	"	DO_3		"		"		"	"
t_{THL2}	"	185	IN	"	OUT				"	"	"	V_{DD}			OUT	OUT	"	10.0 V	DB_0		56		74		44	"
	"	186	GND	IN		OUT			"	"	"	"		OUT			"	"	DB_1		"		"		"	"
	"	187	"	GND			OUT		IN	"	"	"	OUT				"	"	DB_2		"		"		"	"
	"	188	"	"				OUT	GND	"	IN	GND				IN	"	"	DB_3		"		"		"	"
	"	189	"	"					"	"	GND	"					"	"	DO_0		"		"		"	"
	"	190	"	"					"	"	"	"			IN		"	"	DO_1		"		"		"	"
	"	191	"	"					"	"	"	"		IN			"	"	DO_2		"		"		"	"
	"	192	"	"					"	"	"	"	IN				"	"	DO_3		"		"		"	"
t_{TLH1}	"	193	IN	"	OUT				"	"	"	V_{DD}			OUT	OUT	"	5.0 V	DB_0		112		156		80	"
	"	194	GND	IN		OUT			"	"	"	"		OUT			"	"	DB_1		"		"		"	"
	"	195	"	GND			OUT		IN	"	"	"	OUT				"	"	DB_2		"		"		"	"
	"	196	"	"				OUT	GND	"	IN	GND				IN	"	"	DB_3		"		"		"	"
	"	197	"	"					"	"	GND	"					"	"	DO_0		"		"		"	"
	"	198	"	"					"	"	"	"			IN		"	"	DO_1		"		"		"	"
	"	199	"	"					"	"	"	"		IN			"	"	DO_2		"		"		"	"
	"	200	"	"					"	"	"	"	IN				"	"	DO_3		"		"		"	"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases E,F Test no.	Terminal conditions and limits 1/															Measured terminal	Test limits						Unit
			DI_0 1	DI_1 2	DO_0 3	DO_1 4	DO_2 5	DO_3 6	DI_2 7	V_{SS} 8	DI_3 9	$\overline{MRD}$ 10	DB_3 11	DB_2 12	DB_1 13	DB_0 14	CS 15	V_{DD} 16		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{TLH2}	3004 Fig.5a	201	IN	GND					GND	GND	GND	V_{DD}				OUT	V_{DD}	10.0 V	DB_0		56		74		44	ns
	"	202	GND	IN					GND	"	"	"			OUT		"	"	DB_1		"		"		"	"
	"	203	"	GND					IN	"	"	"		OUT			"	"	DB_2		"		"		"	"
	"	204	"	"					GND	"	IN	"	OUT				"	"	DB_3		"		"		"	"
	"	205	"	"	OUT				"	"	GND	GND				IN	"	"	DO_0		"		"		"	"
	"	206	"	"		OUT			"	"	"	"			IN		"	"	DO_1		"		"		"	"
	"	207	"	"			OUT		"	"	"	"		IN			"	"	DO_2		"		"		"	"
	"	208	"	"				OUT	"	"	"	"	IN				"	"	DO_3		"		"		"	"

1/ Pins not designated may be high level logic, low level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, DO and DB terminals shall be open.

2/ Leakage current tests (I_{SS}, I_{IL}, I_{IH}, and I_Z) subgroup 3, are not required for 100 percent screening (final electricals).

3/ Perform a functional test using the test pattern and expected outputs shown below. Input levels shall be 0 = V_{SS} and 1 = V_{DD}. Output levels shall be V_{OL} = V_{SS} +0.50 V max and V_{OH} = V_{DD} -0.5 V min.

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DEVICE TYPE 01

	Name	Pin no.	TIME SLOT: 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
INPUT	CS	15	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	MRD	10	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	DI_0	1	1	1	1	1	0	1	0	0	0	1	1	1	0	1	1	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
	DI_1	2	1	1	1	1	0	0	1	0	0	1	1	0	1	1	0	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
	DI_2	7	1	1	1	1	0	0	0	1	0	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
	DI_3	9	1	1	1	1	0	0	0	0	1	0	1	1	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
			INPUT			OUTPUT																	INPUT														
BIDIRECTIONAL	DB_0	14	0	0	0	1	0	1	0	0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	0	1	1	1	0	1	1	0	1	0	0	1
	DB_1	13	0	0	0	1	0	0	1	0	0	1	1	0	1	1	0	1	0	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0	1	0	1
	DB_2	12	0	0	0	1	0	0	0	1	0	1	0	1	1	0	1	1	0	0	1	1	0	0	1	0	1	0	1	1	0	1	1	0	0	1	1
	DB_3	11	0	0	0	1	0	0	0	0	1	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	1	1	1	0	0	0	1	1	1	1
OUTPUT	DO_0	3	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	1	0	0	0	1	1	1	0	1	1	0	1	0	0	1
	DO_1	4	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	1	0	0	1	1	0	1	1	0	1	0	1	0	1
	DO_2	5	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	0	1	0	1	0	1	1	0	1	1	0	0	1	1
	DO_3	6	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	0	0	1	0	1	1	1	0	0	0	1	1	1	1

DEVICE TYPE 02

	Name	Pin no.	TIME SLOT: 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
INPUT	CS	15	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	MRD	10	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	DI_0	1	1	1	1	1	0	1	0	0	0	1	1	1	0	1	1	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
	DI_1	2	1	1	1	1	0	0	1	0	0	1	1	0	1	1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
	DI_2	7	1	1	1	1	0	0	0	0	0	1	0	1	1	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
	DI_3	9	1	1	1	1	0	0	0	0	1	0	1	1	1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
			INPUT			OUTPUT																	INPUT														
BIDIRECTIONAL	DB_0	14	0	0	0	1	0	1	0	0	0	1	1	1	0	1	1	1	0	0	0	0	0	1	1	1	0	0	0	1	0	0	1	0	1	1	1
	DB_1	13	0	0	0	1	0	0	1	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	1	1	0	0	1	0	0	1	0	1	0	1	1
	DB_2	12	0	0	0	1	0	0	0	1	0	1	0	1	1	0	1	0	1	1	0	0	1	1	0	1	0	1	0	0	1	0	0	1	1	0	1
	DB_3	11	0	0	0	1	0	0	0	0	1	0	1	1	1	0	0	1	0	1	1	0	1	1	1	0	1	0	0	0	1	1	1	0	0	0	1
OUTPUT	DO_0	3	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	0	1	1	1	0	0	0	1	0	0	1	0	1	1	1
	DO_1	4	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	1	0	1	1	0	0	1	0	0	1	0	1	0	1	1
	DO_2	5	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	1	1	0	1	0	1	0	0	1	0	0	1	1	0	1
	DO_3	6	X	X	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0	1	1	1	0	1	0	0	0	1	1	1	0	0	0	1

X = DENOTES HIGH IMPEDANCE STATE.

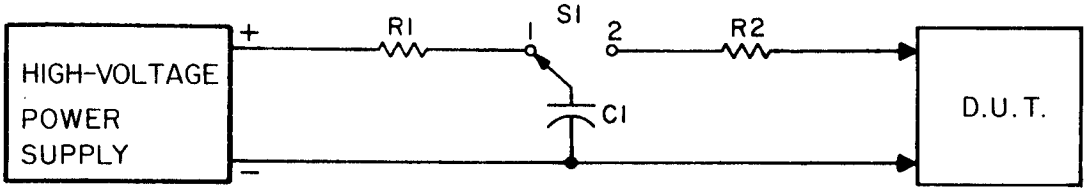
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4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 High voltage (V_{ZAP}) test of input protection circuits. Unless otherwise specified, all input terminals (up to a maximum of four) of the device under test (DUT), shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 6.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C and V_{DD} = 13 V. Also measure I_{SS} at 25°C and V_{DD} = 13 V. These measurements shall be made in accordance with table III. The test limit for a single terminal measurement of I_{IL} and I_{IH} shall be 500 nA maximum.



NOTES:
1. V_{ZAP} = 400 V minimum charge on C1
   10 mΩ ≤ R1 ≤ 50 M.
2. R2 = 1.5 kΩ.
3. C1 = 100 pF
4. S1 = Hg-wetter "bounceless" relay.

FIGURE 6. High voltage (V_{ZAP}) test circuit.

b. With S1 in position 1, charge C1 to 400 V. With S1 in position 2 apply the test voltage (V_{ZAP}) to the same termial selected for leakage current measurements. Apply V_{ZAP} in a three pulse sequence as follows:

(1) Input (-) to V_{DD}
(2) Input (+) to V_{SS}
(3) Input (+) to output

c. Within 24 hours repeat the I_{SS}, I_{IL} and I_{IH} measurements on the same terminals as performed above. At this time a DUT exhibiting leakage currents in excess of the specified limits is defective.

4.6 Data reporting. When specified in the acquisition document, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all dynamic burn-in, and steady state life tests (see 4.4.3).

b. The quality conformance inspection data (see 4.4).

c. Final electrical parameters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. These devices require electrostatic protection.

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6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground. Zero voltage potential.
T_A	Free air temperature.
t_{THL}	Fall time. Time duration which the amplitude of the trailing edge of the input forcing condition or waveform is decreasing from 90 to 10 percent of the maximum amplitude.
t_{TLH}	Rise time. Time duration during which the amplitude of the leading edge of the input forcing condition or waveform is increasing from 10 to 90 percent of the maximum amplitude.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
I_{SS}	Quiescent supply current.

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

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6.7 Substitutability. The cross-reference information herein is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	1856
02	1857

6.8 Handling. CMOS devices should be handled with certain precautions to avoid damage due to accumulation of static charge. Input protection devices have been designed in the ship to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive carriers or conductive foam.

e. Avoid use of plastic, rubber, or silk in areas where CMOS devices are handled.

f. Maintain relative humidity above 50 percent, if practical.

6.9 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
NASA - NA

Review activity:
DLA - ES

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-K646-5)

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